Adriaan Musschoot

Gameplay Developer

- ✗ Foundation in C++
- ✗ Unreal Engine 5
- ✗ Link to portfolio: adriaanmusschoot.com

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https://github.com/AdriaanMusschoot/

Education

Bachelor: Game Development — Graduated Magna Cum Laude — Kortrijk, Belgium
Howest University of Applied Sciences — 2022 – 2025

Work Experience

Black Forest Games Gmbh. — Offenburg, Germany
C++ Programmer Intern — Feb – Jun 2025

- Collaborated with the team on a double-AA Open World Adventure Title in **Unreal Engine 5** using **C++**.
- Developed a **Custom Perception System** for Designers using Unreal Engine's **Mass** framework.
- Improved their **Chain Reaction System** after noticing some shortcomings to **increase usability for designers**.

Games

GetCooked! — The Rookies & GDWC Finalist — Sep – Dec 2024

- Worked in a team of 6 to develop a local pvp game in **Unreal Engine 5** using **C++**.
- Implemented the **Combat System** using **State Machines**, **Player Logic** (including Local Multiplayer) and **localization**.

Snails & Potions — Steam Release & Game Jam Winner — Feb - Jun 2024

- Worked in a team of 5 to develop a party game in **Unreal Engine 5** using **C++**.
- Implemented the **snail's behaviour**, the **brewing mechanics** and the **UI**.
- Finalise the game and integrate **SteamSDK** for the **Steam Release**.

Projects

Raymarcher — C++ | CMake | SDL2 | ImGui — Sep – Oct 2024

- Implemented a software **multi-threaded sphere tracing algorithm**.
- **Optimize rendering** by implementing **Bounding Volume Hierarchy** and **Bounding Box**.
- Integrated **ImGui** to facilitate different environments for **performance-testing**.
- Wrote **research paper**: "Implementing acceleration structures to optimize a sphere tracing ray marcher".

Amugen — C++ | CMake | SDL2 — Feb – Sep 2024

- Designed and created a **custom 2D game engine**, using **C++** and **SDL2**, based on the **Game Object – Component system**.
- Applied **software engineering principles** and **design patterns** such as **RAII**, **Observer** and **Singleton** to provide **scalable** systems for scene management, rendering, input handling, and events.
- **Sound** is handled on a **separate thread** to ensure a smooth game thread.
- Recreated Pac-Man and generated Tectonic puzzles to validate the capabilities of the engine.

Instanced Rendering — C++ | CMake | Vulkan | RenderDoc — Feb – Jun 2024

- Created an application in **Vulkan** using it's **C++** libraries that allows for **instanced rendering of 2D and 3D**.
- Recreated realistic visuals through an input-responsive camera system, using normals, lighting and diffuse textures.

Additional

Technical Skills

- **C++**, C#, CMake, GLSL, JS, HTML, css
- **Unreal Engine 5**, Unity 5
- **Git, Perforce**
- **Mass (UE), SDL2**, Vulkan, ImGui, SteamWorks
- VS2022, RenderDoc, Rider, VSCode, QT

Currently Working On

- GetCooked! Development & Steam Release
- Reading Hands-On Design Patterns With C++ by Fedor G. Pikus
- Web-game in Unity using C#

Languages

- Dutch: Native
- English: C2
- French: B1
- German: A2